

	Ternary	Quaternary	Quinary
Well explored	O-Sr-V	Bi-Cu-Pb-S	C-H-N-O-S
	Mn-O-Se	As-Cl-O-Pb	Ba-Ca-Cu-O-Tl
	La-Mo-O	Fe-Na-O-P	Eu-F-K-O-Si
Partially explored	Li-Pr-Te	Cl-Cu-Dy-Rb	Cu-Gd-O-Ru-Sr
	C-Pr-Ru	Na-Te-Tm-Zr	La-Na-O-Sb-Sc
	C-Mg-Sc	F-Mg-Rb-Sn	Cs-F-O-Tl-Zr
Not explored	Br-Pb-Rh	Cr-Ga-Mg-P	Al-C-H-Sb-Zr
	As-Cu-Sr	C-Cl-Ho-Ru	As-Br-Cr-I-Pt
	Cl-Er-In	Al-Au-Co-S	K-Mo-O-P-Sr

Table D3: Categorization of the 27 chemical systems used to benchmark model capabilities on chemical system exploration

MLFF (MatterSim, see Appendix D.1.4), and the resulting 300,000 MLFF relaxation trajectories were used in the second round of RSS to automatically tune the MINSEP parameter. Again, `airss` was run without structural relaxation followed by a MLFF relaxation. Finally, we combined the 600,000 MLFF-relaxed structures from both rounds and ran DFT structural relaxation and static calculation on the 100 unique structures with the lowest predicted energy above hull according to the MLFF.

Substitution details

A total of 5,143 ordered crystal structures (2,695 ternary, 1,875 quaternary, and 573 quinary) with less than 100 atoms in a unit cell from the ICSD [92] were used as prototypes. For each chemical system in Table D3, we computed all possible unique substitutions of the prototypes, relaxed all structures using a MLFF (MatterSim, see Appendix D.1.4), and selected the 100 unique structures with the lowest predicted energy above the hull according to the MLFF. Finally, we ran DFT structural relaxation and static calculation on the selected structures.

D.4.2 Additional qualitative analysis of structures

The V-Sr-O chemical system example provided in Fig. 3 produced four new on-hull crystal structures: SrV_2O_6 (V^{5+}), SrVO_3 (V^{4+}), $\text{Sr}_3\text{V}_2\text{O}_8$ (V^{5+}) and SrV_2O_4 (V^{4+}). This chemical system has been well-studied in literature, with SrVO_3 being a well-known perovskite [93], Sr_2VO_4 expected to crystallize into a K_2NiF_6 -like crystal structure, and $\text{Sr}_3(\text{VO}_4)_2$ synthesized in a cation-deficient variant of the SrVO_3 crystal structure [94].

Vanadates are known to be synthesizable in a variety of frameworks, with the expected co-ordination of the VO_4 sub-unit varying with oxidation state [95] from the ideal tetrahedron in V^{5+} to a variety of other co-ordination environments. All generated structures have plausible atomic environments with VO_4 sub-units, either ideal or distorted, and oxygen co-ordinated Sr atoms.

One SrV_2O_4 structure, having $\text{P}\bar{1}$ symmetry, consists of a layers of ideal VO_4 edge-sharing tetrahedra separated by a Sr in a triangular prismatic bonding configuration, resulting in a 1D channel of voids in the Sr layer.

D.5 Designing materials with target symmetry

This section provides supplementary information to the results in Section 2.4.

D.5.1 Additional experimental details

For generating structures belonging to a target space group, we fine-tune our base model on the whole training set, and represent the latent embedding of the space group of a crystal via one-hot encoding of the space group.

We assess the capability of our model to correctly generate structures belonging to any space group via two tasks. For the first task (Fig. 4), we sample two space groups for each of the seven lattice systems, and choose to sample only from space groups that contain at least 1000 structures in the training set. We then compute the fraction of S.U.N. structures our fine-tuned model generates when conditioned on these space groups that are classified as belonging to that space group according to the `SpaceGroupAnalyzer` module of `pymatgen` [79, 96]. This metric is computed for 256 generated structures per space group after DFT relaxation has been performed. For the second task (Fig. D4), we generate 10,000 structures conditioned on space groups sampled randomly from the data distribution of the training set, and check whether our model is able to reproduce the distribution of space groups from the training data. For both of the above tasks, the number of atoms in the systems are sampled from the distribution of number of atoms for that space group in the training set. This way, we avoid ‘impossible’ space group constraints, where the space group we condition on cannot be satisfied given the number of atoms we set.

D.5.2 Additional Analysis of Structures

The highlighted examples in Fig. 4 show a high degree of novelty, with few matching known prototypes. The combination of elements in generated structures is also not common, e.g., DyScNiPd, CeAsRh. It is therefore difficult to reference to known structures in literature. Half of the examples could be assigned formal valences, and the majority were robust for their target symmetry even when symmetry was calculated with a higher symmetry-finding tolerance.

D.6 Designing materials with target magnetic, electronic and mechanical properties

This section provides supplementary information to the results in Section 2.5.

D.6.1 Additional experimental details

To generate structures conditioned on a target property, we fine-tune our base model on magnetic density ($N \approx 605,000$ DFT labels), band gap ($N \approx 42,000$) and bulk modulus ($N \approx 5,000$), respectively. See Appendix B.1 for more details on the fine-tuning scheme, and Appendix D.1.2 for hyperparameter settings. We represent the latent embedding of scalar properties via a sinusoidal encoding from the Transformer architecture [97].

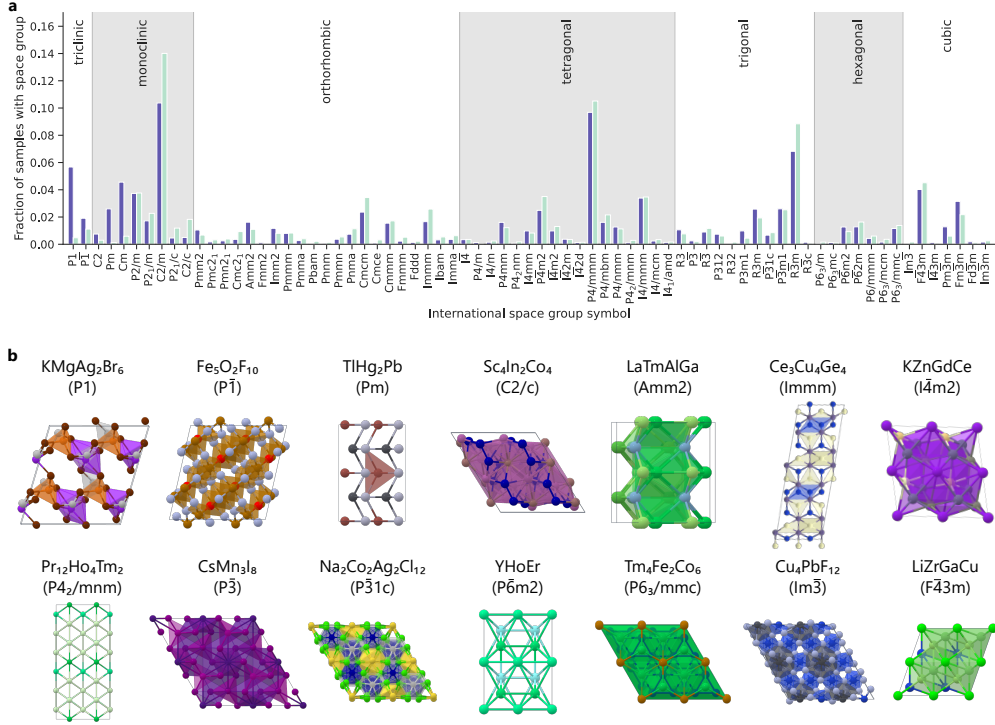


Fig. D4: Generating materials with target symmetry. (a) Fraction of generated structures belonging to space groups generated by MatterGen fine-tuned on symmetry groups, sampled unconditionally (violet), and structures present in the reference dataset (green). (b) Fourteen generated S.U.N. structures, one for each space group reported in Fig. 4; composition and space group are reported.

For each property in Fig. 5(a-c), we generate 512 samples with our fine-tuned model by conditioning on a value of $0.2 \text{ \AA}^{-3}$ for magnetic density, 3.0 eV for band gap, and 400 GPa for bulk modulus. We relax those structures using the MLFF and filter the relaxed structures by stability and uniqueness. We then relax the remaining structures with DFT and filter by stability, uniqueness, and novelty. Finally, we compute the desired property of the remaining structures using DFT, and filter out structures where we consider the computed property to be an outlier, i.e., high bulk modulus values with more than 600 GPa and magnetic density values with more than $0.3 \text{ \AA}^{-3}$. For more details on the DFT calculations, see Appendix C.2. After all filters and DFT computations, we obtain 251 S.U.N. structures for magnetic density, 142 for band gap, and 22 for bulk modulus.

For Fig. 5(g), we use MatterGen to generate 15,360 samples conditioned on a magnetic density value of $0.2 \text{ \AA}^{-3}$, relax those structures using the MLFF and filter the relaxed structures by stability and uniqueness. This results in 5,365 candidate structures. Next, we randomly sub-sample 600 structures and relax them via DFT, with

540 of them being DFT stable. We then randomly sub-sample k structures according to the given DFT property calculation budget, and compute their magnetic density via DFT. As a baseline, we count the number of structures in the labeled training dataset that satisfy the target property constraint. For Fig. 5(h), we train a separate property predictor for bulk modulus (see details below), which we use for both MatterGen and the screening baseline. In particular, we use predicted bulk modulus values to fine-tune the base MatterGen model. We generate 8,192 samples conditioned on a bulk modulus value of 400 GPa, relax those structures using the MLFF and filter the relaxed structures by stability and uniqueness. This results in 801 candidate structures. We relax those via DFT, with 736 of them being DFT stable. We then randomly sub-sample k structures according to the given DFT property calculation budget, and compute their bulk modulus via DFT. For the screening baseline, we use the bulk modulus property predictor to predict the bulk modulus values of all structures in the training dataset for which we do not have an existing DFT label, rank those structures by their predicted bulk modulus values, and choose the top k structures according the DFT property calculation budget. We then verify their actual bulk modulus values via DFT.

Property predictor details

The bulk modulus property predictor used in Fig. 5(h) consists of a GemNet-dT [70] encoder that provides atom and edge embeddings, followed by a mean readout layer. We employ three message-passing layers, a cutoff radius of 10 Å for the neighbor list construction, and set the dimension of nodes and edges hidden representations to 128.

To train the property predictor, we use all materials with DFT Voigt-Reuss-Hill average bulk modulus values from MP [15] (including structures with more than 20 atoms), which are 7,108 structures in total. We allocate 80% of the data for the training set, 10% for validation, and 10% for testing. We follow the MatBench benchmark [98] and predict the $\log_{10}$ bulk modulus. At the end of training, the model achieves a mean absolute error (MAE) of 9.5 GPa. The model was trained using the Adam optimizer. Gradient clipping was applied by value at 0.5. The learning rate was initialized at 5×10^{-4} and decayed using the ReduceLROnPlateau scheduler with decay factor 0.8, patience 10 and minimum learning rate 10^{-8} . The training was stopped when the validation loss stopped improving for 150 epochs.

D.6.2 Additional qualitative analysis of structures

Magnetic density conditional generation

For the high magnetic density generation task, a manual review of ten generated crystals chosen at random was performed, as well as the two representative structures with high magnetization density that were highlighted in Fig. 5. Of the random selection, eight of the ten generated structures were ordered approximations of a Fe-Co alloy: α -Fe with a partial Co substitution from 10% Co to 40% Co. The Fe-Co system is a well-known and versatile soft magnetic material system with a wide region where a α -Fe_xCo_{1-x} phase is stable, with a mixture of γ -Fe and α -Co expected at lower Co content. While these generated structures could be considered “good”, in the sense that they are physically plausible and indeed would be useful magnetic materials, they

are likely only considered novel since these specific ordered unit cells do not currently exist in the reference databases. Also, as an alloy, predicted quantities such as energy above hull based on a single ordered approximation will be misleading. Two of the ten generated structures contained hydrogen, either as Fe_9H_2 or $\text{Fe}_4\text{Co}_2\text{H}$, with hydrogen in an ideal octahedral environment. The generation of these structures can be rationalized owing to extensive study of iron hydrides, with the generated structures having local atomic environments similar to that in a double hexagonal close-packed iron hydride structure that can exist at high pressure. The realization of these specific structures is unlikely with phase segregation expected to occur. The two structures highlighted in Fig. 5(d) contain Gd, an element with a large magnetic moment due to its 7 unpaired electrons when in its Gd^{3+} oxidation state and in its elemental state. Therefore, it is unsurprising that the model would preferentially generate materials containing Gd in this instance: as in the randomly selected sample, this task necessarily shows a strong compositional bias in the materials generated. The Gd_2N example is a layered material. Although the Gd_2N molecule has been studied [99], it is unknown if and how it might crystallize. The generated $\text{Gd}_6\text{H}_2\text{CN}_3$ structure is rocksalt-derived, with Gd on one site, and a mix of C, H and N on another site, with all atoms in almost ideal octahedral environments.

Band gap conditional generation

For the target band gap generation task, a manual review of ten generated crystals chosen at random was performed, as well as the two representative structures with desired target band gap that were highlighted in Fig. 5. Unlike the other single property optimization tasks, the generated crystals did not show a strong compositional dependence, with a wide range of elements presented in generated materials. All generated materials also could nominally charge balance, unlike the materials analyzed from other single property optimization tasks. This seems reasonable for a task designed to generate insulating systems; the target band gap of 3 eV (calculated with the PBE functional, and thus underestimating the true electronic band gap) should generate insulating materials, and thus more ionic solids. Of the random sample, we could only find NaNO_3 and the molecular crystal BI_3 as compositions that had previously been synthesized. Both were in incorrect symmetry compared to their experimentally known structures, with the generated NaNO_3 in $\text{C}2/c$ compared to the experimental $\text{R}\bar{3}c$, and BI_3 in $\text{P}1$ compared to the experimental $\text{P}6_3/m$. However, both had correct local bonding and similar calculated band gaps to the band gaps calculated for their experimental structures. This indicates one instance whereby the model might still generate useful results even if the generated crystal structure is different from the experimental ground state, if it can guide a scientist towards investigation of a specific system.

The examples highlighted in Fig. 5(e) were VBiO_4 and TlNO_3 . Of these, the local bonding in VBiO_4 was very similar to that in the experimentally-known bismuth vanadate which crystallizes in the $\text{I}4_1/\text{amd}$ [100] space group or $\text{I}4_1/a$ space group [101], unlike the generated $\text{P}2_1/m$ space group, and so is another example of reasonable local environment but with a seemingly incorrect space group. Nevertheless, this crystal structure was calculated to be thermodynamically stable with respect to these experimentally-known crystal structures; this could be a limitation with respect to the